

Teorema de Ehrenfest

$$\frac{d}{dt} A_\psi = -\frac{i}{\hbar} [A, H]_\psi + \left(\frac{\partial A}{\partial t}\right)_\psi$$

Sea A un operador lineal, la derivada temporal de su valor esperado es:

$$\frac{d}{dt} (\psi, A\psi) = \frac{d}{dt} \int \psi^* A\psi d^3\vec{r} = \left(\frac{\partial \psi}{\partial t}, A\psi\right) + (\psi, A\frac{\partial \psi}{\partial t}) + (\psi, \frac{\partial A}{\partial t}\psi)$$

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi \rightarrow \frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} H\psi$$

$$\begin{aligned} \frac{d}{dt} (\psi, A\psi) - (\psi, \frac{\partial A}{\partial t}\psi) &= \left(\frac{\partial \psi}{\partial t}, A\psi\right) + (\psi, A\frac{\partial \psi}{\partial t}) \\ &= \frac{i}{\hbar} (H\psi, A\psi) - \frac{i}{\hbar} (\psi, AH\psi) \\ &= -\frac{i}{\hbar} (\psi, [A, H]\psi) \end{aligned}$$

$$\boxed{\frac{d}{dt} A_\psi = -\frac{i}{\hbar} [A, H]_\psi + \left(\frac{\partial A}{\partial t}\right)_\psi}$$

Ejm

$$1) [X, H] = [X, \frac{P^2}{2m} + V] = [X, \frac{P^2}{2m}] = i\frac{P\hbar}{m}$$

$$\frac{d}{dt} X_\psi = \frac{P_\psi}{m}$$

$$2) [P, H] = [P, \frac{P^2}{2m} + V] = [P, V] = -i\hbar \nabla V$$

$$\boxed{[X, P] = i\hbar}$$

$$\boxed{[X, P^2] = \frac{P}{2m} [X, P] + [X, P] \frac{P}{2m}}$$

$$\begin{aligned} [P, V]\psi &= P(V\psi) - V P\psi \\ &= \underline{P} V \psi + \underline{V P} \psi - \underline{V P} \psi \end{aligned}$$

$$\frac{d}{dt} P_{\psi} = - \nabla V = F$$

Si el operador H no depende explícitamente del tiempo,
 como $H = \frac{p^2}{2m} + V(R)$, entonces:

$$\frac{d}{dt} H_{\psi} = - \frac{i}{\hbar} [H, H]_{\psi} \overset{\frac{\partial H}{\partial t} = 0}{=} 0 \rightarrow H_{\psi} = \text{cte}$$

Ejm $P_{\psi} = m \dot{X}_{\psi} \rightarrow \dot{P}_{\psi} = m \ddot{X}_{\psi}$

$$\dot{P}_{\psi} = - \left(\frac{\partial V}{\partial x} \right)_{\psi}$$

$$m \ddot{X}_{\psi} = - \left(\frac{\partial V}{\partial x} \right)_{\psi} = -K X_{\psi} \rightarrow \ddot{X}_{\psi} = - \frac{K}{m} X_{\psi}$$

$$X_{\psi} = A \sin \omega t + B \cos \omega t$$

$$\dot{X}_{\psi} = A\omega \cos \omega t - B\omega \sin \omega t$$

$$\ddot{X}_{\psi} = -A\omega^2 \sin \omega t - B\omega^2 \cos \omega t$$

$$= -\omega^2 X_{\psi}, \quad \omega = \sqrt{\frac{k}{m}}$$

NOTA

$$[X, P] = i\hbar \rightarrow XP\psi - P(X\psi) = i\hbar \psi$$

$$\underline{XP\psi} - (PX)\psi - \underline{XP\psi} = i\hbar \psi$$

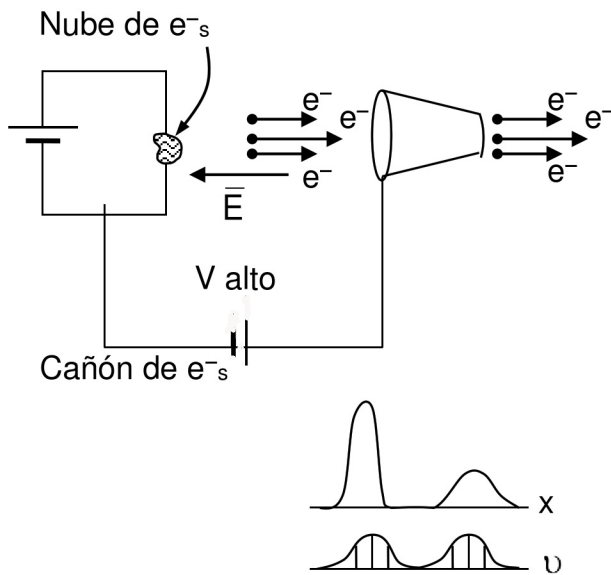
$$-(PX)\psi = i\hbar \psi \quad P \equiv \frac{\hbar}{i} \frac{\partial}{\partial x}$$

$$[X, P]\tilde{\psi} = i\hbar \tilde{\psi} \rightarrow X(P\tilde{\psi}) - P(X\tilde{\psi}) = i\hbar \tilde{\psi}$$

$$(XP)\tilde{\psi} + \underline{PX\tilde{\psi}} - \underline{PX\tilde{\psi}} = i\hbar \tilde{\psi}$$

$$X \equiv -\frac{\hbar}{i} \frac{\partial}{\partial P}$$

Partícula Libre



Partícula Libre:

- 1) Distribución gaussiana de posición y momentum en t.
- 2) Avance con $V = \text{cte}$ de R_ψ es acompañado del ensanchamiento de la curva de distribución de posición.
- 3) Inalterabilidad de la curva de distribución de momentums.

Hipótesis de Equivalencia:

Haz de N electrones	$\leftrightarrow$	e^- haciendo el recorrido N veces
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$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{\hbar}{i} \partial_t \psi = 0$$

$$\psi_t(\vec{r}) = e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\psi_t(\vec{r}) = c \int d^3 \vec{x} \phi(\vec{k}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

Coef. Fourier $\omega = \frac{\hbar k^2}{2m}$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int d^3\vec{r} \psi(0, \vec{r}) e^{-i\vec{k} \cdot \vec{r}}$$

En 1D:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int dk \phi(k) e^{i(xk - \omega t)}$$

$$\hbar k$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int dx \psi(0, x) e^{-ikx}$$

Recuerde que:

$$v(x) = e^{-ax^2}$$

$$\phi(k) = \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-ikx} \left(\frac{1}{\sqrt{2\pi}} \right)$$

$$-ax^2 - ikx = -a \left(x^2 + \frac{ik}{a} x \right)$$

$$= -a \left[\left(x + \frac{ik}{2a} \right)^2 + \frac{k^2}{4a^2} \right]$$

$$\phi(k) = e^{-\frac{k^2}{4a}} \underbrace{\int dx e^{-a \left(x + \frac{ik}{2a} \right)^2}}_{\sqrt{\frac{\pi}{a}}}$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} e^{-\frac{k^2}{4a}}$$

$$P_v(x) = \frac{\hbar}{i} \frac{d}{dx} e^{-ax^2} = \frac{\hbar}{i} (-2ax) e^{-ax^2}$$

$$P_\psi = \int_{-\infty}^{\infty} v^* P_v dx = -\frac{2a\hbar}{i} \int_{-\infty}^{\infty} x e^{-2ax^2} dx = 0$$

Esto indica que $P_v = 0$, lo que quiere decir que  (no representa a una partícula en movimiento)

$$\begin{aligned} n(t, \vec{r}) &= e^{i(\vec{k} \cdot \vec{r} - \omega_k t)} \\ v(x) &= e^{-ax^2} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{no son} \\ \text{compatibles} \end{array}$$

P_ψ, R_ψ son reales

$$\eta(x) = \underline{A(x)} e^{ikx} \rightarrow \frac{d\eta}{dx} = A' e^{ikx} + ikA e^{ikx}$$

$$P_\psi = \frac{\hbar}{i} \int \eta^* \frac{d\eta}{dx} dx = \frac{\hbar}{i} \int_{-\infty}^{\infty} A^* A' dx + \frac{\hbar}{i} ik \int_{-\infty}^{\infty} A^* A dx$$

Se debe elegir A de modo que:

$$\int dx A^* A' = 0$$

$$A = e^{-ax^2}$$

Tomaremos

$$\psi(x) = B e^{-ax^2 + ibx}$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int dx \psi(x) e^{-ikx}$$

$$\phi(k) = \frac{B}{\sqrt{2\pi}} \int dx e^{-ax^2 + i(b-k)x}$$

$$-ax^2 + i(b-k)x = -a \left[\left(x - \frac{i(b-k)}{2a} \right)^2 + \frac{(b-k)^2}{4a^2} \right]$$

$$\phi(k) = B \frac{e^{-\frac{(b-k)^2}{4a}}}{\sqrt{2\pi}} \int dx e^{-a \left(x - \frac{i(b-k)}{2a} \right)^2}$$

$$\rightarrow \phi(k) = \frac{B}{\sqrt{2a}} e^{-\frac{(b-k)^2}{4a}}$$

Paquete de onda $\equiv$ partícula libre

$$\psi(t, x) = \frac{1}{\sqrt{2\pi}} \int dk \phi(k) e^{i(kx - \omega_k t)}; \quad \omega_k = \frac{\hbar k^2}{2m}$$

Argumento exponente

$$-\frac{(k-b)^2}{4a} + i\left(kx - \frac{\hbar k^2}{2m}t\right)$$

$$= -\frac{N}{4a}\left[k - \frac{b+2ia}{N}x\right]^2 - \frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}$$

$$N = 1 + i\alpha$$

$$\alpha = 2\frac{\hbar a}{m}t$$

$$\psi(t, x) = \frac{B}{N^{1/2}} e^{-\frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}}$$

$$-\frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}$$

$$= \left(-\frac{a}{|N|^2}\right) \left[\left(x - \frac{\alpha b}{2a}\right)^2 + i\left(\frac{a}{|N|^2}\right) \cdot \left[\alpha\left(x^2 - \frac{b^2}{4a^2}\right) + \frac{b}{a}x \right] \right]$$

$$\rho(t, x) = |\psi(t, x)|^2$$

$$= \frac{B^2}{|N|} e^{-\frac{2a}{|N|^2}\left(x - \frac{\hbar b t}{m}\right)^2}$$

$$R_\psi = (\psi, R\psi) = \frac{\hbar b t}{m} \quad \begin{array}{l} \text{(por 1er teorema de} \\ \text{Ehrenfest,} \\ \text{encuentras } P_\psi) \end{array}$$

$$\rightarrow P_\psi = (\psi, P\psi) = \hbar b$$

$$\sigma_\psi R = \frac{|N|}{2\sqrt{a}} \quad N = 1 + i\alpha t$$

$$\sigma_\psi P = \sqrt{a} \hbar \quad \xi = \frac{2\hbar a}{m}$$

$$\sigma_\psi R \sigma_\psi P = \frac{|N| \hbar}{2}$$

(cumple principio de incertidumbre Heisenberg)

$$\frac{1}{\sqrt{2\pi}} \frac{B}{\sqrt{2a}} \sqrt{\frac{\pi}{N}} \sqrt{4a} = \frac{B}{N^{1/2}}$$